A Multilingual Stroop Benchmark for Testing Visual-Textual Cue Interference in Vision-Language Models

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Abstract*—Vision-language models are often evaluated on image captioning, retrieval, or visual question answering benchmarks, but these tasks may not isolate whether the model follows visual evidence or textual semantic shortcuts when both are present in the same image. This paper presents an expanded synthetic Stroop-style benchmark for studying visual-textual cue interference in vision-language models. The benchmark asks a deliberately simple question: when a word is printed in a particular font color, does the model identify the visible font color, or is its decision pulled toward the written word meaning? The full experimental pool contains 140,000 generated images across a 40,000-image English Stroop set, a 20,000-image multilingual text-meaning control, and an 80,000-image multilingual Stroop set covering English, Chinese, Hindi, and Arabic. CLIP ViT-B/16 is evaluated using zero-shot image-text similarity, while Qwen2.5-VL-3B is evaluated as an instruction-following vision-language model on a balanced 16,000-image subset. The CLIP results show severe English Stroop-like word bias: on the main English set, CLIP predicts the written word rather than the visible font color for 91.72% of incongruent images and 88.96% of background-conflict images. Prompt robustness experiments over 12 templates show that longer and more explicit prompts reduce but do not remove the effect. The multilingual results show that CLIP's word bias is high for English but remains very low for Chinese, Hindi, and Arabic, where errors shift mainly toward background color or other visual confusions. The text-meaning control further supports this interpretation: CLIP recognizes English words perfectly in the clean black-text condition but performs poorly for the non-English scripts. Qwen2.5-VL-3B behaves differently, achieving high font-color accuracy and low word bias on most conditions, although background-conflict cases remain harder. Finally, CLIP Grad-CAM-style heatmaps and Qwen attention heatmaps are included as qualitative interpretability analyses. The results suggest that Stroop-style benchmarks are useful for testing whether multimodal models can separate what they read from what they see.

***Index Terms*—vision-language models, CLIP, Qwen2.5-VL, Stroop effect, prompt robustness, multilingual evaluation, interpretability, Grad-CAM, attention heatmaps**

I. Introduction

Vision-language models (VLMs) connect visual inputs with language. In many applications, these models must answer questions about images that contain both visual attributes and readable text. This creates a subtle but important problem: the model may answer using the semantic meaning of text instead of using the requested visual attribute. For example, if the word `RED` is printed in yellow, the correct answer to the question "What is the font color?" is yellow. A model that answers red has not simply made a random classification error; it has followed the written word cue instead of the visible color cue.

This paper studies this behavior using a synthetic Stroop-style benchmark. The human Stroop effect shows that word meaning can interfere with color naming when the semantic word and the ink color conflict (Stroop, 1935). Here, the same idea is used as a controlled test for VLMs. The goal is not to claim that VLMs experience human cognition, but to use a Stroop-style conflict as a diagnostic tool for visual-semantic binding.

The central question is: can a VLM separate what it reads from what it sees? This question matters beyond toy color images. In real systems, a model may be asked to inspect a warning label, a medical chart, a user interface, a road sign, or a document where text and visual cues can disagree. A robust model should follow the requested cue rather than automatically privileging the most semantically salient cue.

An earlier stage of this project used a 20,000-image English-only CLIP benchmark and custom CLIP white-box attribution. That stage showed strong word bias in CLIP under English color-word conflict. The present paper expands the work into a more systematic benchmark with explicit dataset ledgers, larger sample sizes, prompt-style ablations, multilingual scripts, Qwen2.5-VL evaluation, and separate interpretability outputs. The previous CLIP white-box work is mentioned only as background continuity; the main focus of this paper is the expanded benchmark and the new results.

The contributions are:

- A 140,000-image synthetic benchmark that separates English Stroop, multilingual Stroop, and text-meaning control experiments.
- A prompt robustness experiment evaluating 12 prompt templates ranging from vague to highly descriptive instructions.
- A multilingual Stroop analysis across English, Chinese, Hindi, and Arabic, with identical semantic color labels but different written scripts.

- A comparison between contrastive CLIP zero-shot classification and instruction-following Qwen2.5-VL-3B on a balanced 16,000-image subset.
- Qualitative interpretability using custom CLIP Grad-CAM-style heatmaps and Qwen answer-token attention heatmaps.

II. Related Work

A. Vision-Language Models and Zero-Shot Recognition

CLIP introduced a contrastive image-text learning framework in which an image encoder and a text encoder are trained to align visual and linguistic representations (Radford et al., 2021). This design enables zero-shot classification by comparing one image against a set of candidate text prompts rather than training a task-specific classifier. Vision Transformers are commonly used as CLIP image encoders, dividing images into patches and processing them through transformer layers (Dosovitskiy et al., 2021). Later work such as OpenCLIP, SigLIP, and BLIP-2 has extended contrastive and multimodal pre-training at larger scale or with different losses and language-model components (Cherti et al., 2023; Zhai et al., 2023; Li et al., 2023). These models provide strong general-purpose recognition capabilities, but their behavior can still depend strongly on how visual cues are represented in prompts.

B. Prompt Learning and Prompt Sensitivity

Prompt design is central to CLIP-style evaluation because labels are represented as natural-language prompts. Prompt-learning methods such as CoOp and CoCoOp show that learned or conditional prompts can improve transfer performance (Zhou et al., 2022a; Zhou et al., 2022b). However, prompt sensitivity also creates an evaluation challenge: poor performance may reflect either a genuine visual failure or an unsuitable prompt. For this reason, this paper does not rely on a single prompt. It evaluates vague, ambiguous, exact, descriptive, and control prompt styles to test whether CLIP's word bias is a prompt artifact or a stable failure mode.

C. Visual Attribute Binding and Color Bias

The task studied here is related to broader work on attribute binding and concept association in VLMs. Tang et al. (2023) show that vision-language models can associate objects with typical attributes even when the image evidence conflicts with those associations. Arias et al. (2025) specifically discuss color-related deficiencies in CLIP-like models, while Samin et al. (2025) examine color understanding failures in large vision-language models. The present benchmark differs by using text-rendered synthetic images where semantic word meaning, font color, and background color are independently controlled. This makes it possible to classify each error as word bias, background bias, or another error rather than reporting only accuracy.

D. Interpretability for VLMs

The interpretability component of this work is connected to Grad-CAM and gradient-based CLIP explanation methods. Grad-CAM uses gradients flowing into convolutional feature maps to localize regions important for a prediction (Selvaraju et al., 2017). Grad-ECLIP extends gradient-based visual and textual explanation to CLIP-style image-text similarity (Zhao et al., 2025). In this project, the CLIP heatmaps are described as custom Grad-CAM-style attributions because they use gradients and activations from CLIP's visual encoder rather than the official Grad-ECLIP implementation. Qwen2.5-VL heatmaps are reported separately as attention-based visual-token diagnostics because the model is instruction-following and autoregressive rather than contrastive (Bai et al., 2025).

III. Methodology

A. Research Requirements and Analysis

The project was designed around four experimental requirements. First, the benchmark must isolate visual font color from written word meaning. This requires synthetic data because natural images rarely provide exact control over word, font color, and background color. Second, the benchmark must distinguish different failure types, not just correct and incorrect predictions. Third, the evaluation must test whether word bias is robust to prompt wording. Fourth, the multilingual extension must preserve the same semantic color labels while changing only the written script, so that cross-script differences can be interpreted as differences in script-level text processing.

The target behavior is straightforward: the correct answer is always the visible font color. A model that predicts the written word color is treated as following a semantic distractor. A model that predicts the background color is treated as following a visual contextual distractor. These requirements led to the four-condition dataset design, the prompt battery, the text-meaning control, and the multilingual Stroop experiment.

B. Benchmark Design

Each Stroop-style image contains a single rendered word, a visible font color, and a background color. The prediction target is always the visible font color:

$$y = \text{font\_color}(x). \tag{1}$$

The written word and the background are treated as distractor cues. The 10-color label space is:

$$\begin{aligned} C = \{&\text{red, blue, green, yellow, orange,} \\ &\text{purple, brown, gray, black, white}\}. \end{aligned} \tag{2}$$

For color-word examples, the word meaning is mapped into the same label space C. For neutral examples, the word does not correspond to a color class.

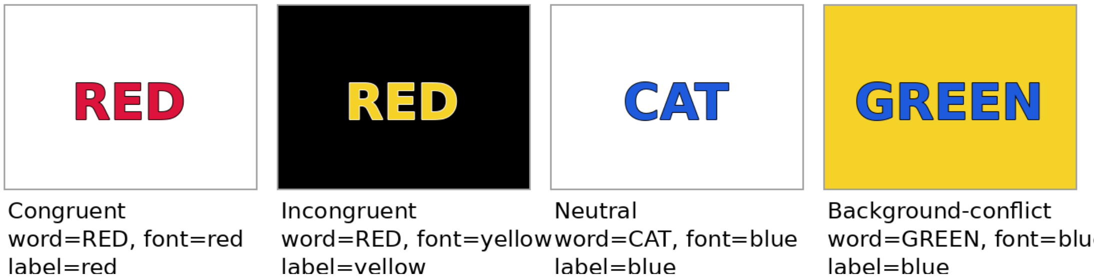

Fig. 1: Illustrative Stroop conditions used in the benchmark. The correct label is always the visible font color, not the written word and not the background.

TABLE I: Dataset ledger for the expanded benchmark.

Experiment	Purpose	Composition	Unique images	Evaluations
Main English Stroop	Test CLIP font-color recognition under English cue conflict	1 language $\times$ 4 conditions $\times$ 10,000	40,000	40,000
Prompt robustness	Test whether word bias changes with prompt specificity	Same 40,000 English images $\times$ 12 prompt templates	40,000	480,000
Text-meaning control	Test clean word recognition without color conflict	4 languages $\times$ 20 word classes $\times$ 250	20,000	20,000 per evaluation mode
Multilingual Stroop	Test script-dependent word bias	4 languages $\times$ 4 conditions $\times$ 5,000	80,000	80,000
Qwen subset	Evaluate instruction-following VLMs with manageable autoregressive cost	4 languages $\times$ 4 conditions $\times$ 1,000	16,000	16,000
Heatmaps	Qualitative interpretability	Selected representative predictions	selected	selected

C. Dataset Generation

Images are generated at 224×224 resolution to match the standard CLIP input size. Each image contains one centered word with small positional jitter. For every sample, the generator saves: (i) the RGB image used for model evaluation, (ii) a binary text mask used only for heatmap analysis, and (iii) metadata containing the language, condition, written word, word-color meaning, font color, background color, and ground-truth label.

The benchmark uses four conditions: **congruent**, where word meaning and font color match; **incongruent**, where word meaning and font color conflict; **neutral**, where the word is not a color term; and **background-conflict**, where word meaning, font color, and background color are all different cues. Fig. 1 gives examples of these cases.

D. Dataset Ledger

A key design goal is to make every experiment traceable by image count. Table I summarizes the full benchmark. The prompt robustness experiment reuses the same 40,000 English images; therefore it increases the number of image-prompt evaluations but not the number of unique images.

E. Model Evaluation Procedure

CLIP learns image and text encoders in a shared embedding space (Radford et al., 2021). Given image x and candidate prompt p_c for color c, CLIP computes normalized embeddings $f_I(x)$ and $f_T(p_c)$. The similarity score is:

$$s(x, p_c) = f_I(x)^\top f_T(p_c). \tag{3}$$

The zero-shot prediction is:

$$\hat{y} = \arg\max_{c \in C} s(x, p_c). \tag{4}$$

For each prompt template, 10 candidate prompts are constructed by replacing the color slot with every color in C. The predicted color is the candidate with the highest image-text similarity. The main CLIP model used in the expanded experiments is CLIP ViT-B/16.

Qwen2.5-VL is evaluated differently because it is an instruction-following VLM (Bai et al., 2025). It receives an image and a natural-language question, then generates a textual answer. The generated answer is parsed using the same 10 canonical color names. Qwen is run on a balanced 16,000-image subset because autoregressive VLM evaluation is much slower than CLIP evaluation.

F. Implementation

The benchmark was implemented in a Google Colab notebook. The implementation includes dataset generation, metadata saving, CLIP zero-shot evaluation, text-meaning evaluation, multilingual evaluation, Qwen evaluation with checkpointing, and heatmap generation. The generator uses Python image rendering utilities and stores one image file, one text-mask file, and one metadata row per sample. Results are saved as CSV files for predictions, summaries, and heatmap metrics. Model outputs are organized into separate CLIP and Qwen result directories so that the dissertation results can be reproduced and inspected independently.

For long-running Qwen experiments, checkpoint files are written to Google Drive. This prevents loss of results if

TABLE II: Prompt styles used in the robustness experiment.

Style	**Purpose**
Vague	Minimal color association, e.g., a bare color word.
Ambiguous	Original wording that may be read as font color or writing/language.
Exact	Directly asks for ink, font, or printed-letter color.
Descriptive	Explicitly tells the model to ignore word meaning and/or background.
Control	Intentionally word-oriented or background-oriented prompts.

Colab disconnects. On restart, the notebook reloads the saved prediction CSV, skips already processed image identifiers, and continues from the remaining images. This is important because evaluating 16,000 autoregressive VLM examples is substantially slower than embedding-based CLIP evaluation.

G. Testing and Evaluation Criteria

Testing is performed through quantitative and qualitative evaluation. Quantitatively, each prediction is categorized as correct font color, word bias, background bias, or other error:

$$\mathrm{error}(x) = \begin{cases} \text{correct\_font\_color}, & \hat{y} = y, \\ \text{word\_bias}, & \hat{y} = \text{word\_color}(x), \\ \text{background\_bias}, & \hat{y} = \text{background\_color}(x), \\ \text{other\_error}, & \text{otherwise.} \end{cases} \tag{5}$$

The main metrics are accuracy, word-bias rate, background-bias rate, and other-error rate. Qualitatively, CLIP Grad-CAM-style heatmaps and Qwen attention heatmaps are used to inspect selected model decisions.

IV. Prompt Robustness Design

Prompt robustness is included because a vague template may unintentionally invite word reading. The expanded benchmark evaluates 12 templates grouped into five styles, summarized in Table II. The main goal is to test whether word bias decreases as the instruction becomes more explicit.

This design separates three possibilities: the effect may be caused by ambiguous wording, it may be reduced but not removed by clearer wording, or it may persist even when the prompt directly specifies the visual font-color task. The results show that the third case is closest to the observed CLIP behavior.

V. Results

This section reports the experiments in the same order as the evaluation pipeline. Each subsection gives the relevant result table or figure immediately after the paragraph that introduces it, so the reader can connect each experiment with its evidence.

A. Experiment 1: Main English Stroop Results

Table III shows the CLIP ViT-B/16 results on the 40,000-image English Stroop dataset. CLIP is perfect on congruent examples but severely biased on conflict examples. In incongruent images, word bias reaches 91.72%. In background-conflict images, word bias remains 88.96%. Neutral accuracy is 79.74%, showing that CLIP can often identify visible font color when the word is not itself a color label.

TABLE III: CLIP ViT-B/16 on the 40,000-image main English Stroop dataset. Values are percentages.

Condition	N	Acc.	Word	Bg.	Other
congruent	10,000	100.00	0.00	0.00	0.00
incongruent	10,000	8.28	91.72	0.00	0.00
neutral	10,000	79.74	0.00	2.37	17.89
background_conflict	10,000	5.23	88.96	5.81	0.00

B. Experiment 2: Prompt Robustness

Table IV and Fig. 2 show the prompt robustness results. The key finding is that prompt wording matters but does not solve the problem. Exact and descriptive prompts reduce word bias slightly, but conflict word-bias rates remain high. For example, the exact ink-color prompt still produces 91.72% word bias on incongruent examples and 88.96% word bias on background-conflict examples. The descriptive prompt that says to ignore the word reduces incongruent word bias to 89.52%, but this remains a severe failure.

TABLE IV: Prompt robustness results for CLIP ViT-B/16 on the 40,000 English Stroop images. Values are percentages.

Prompt key	Style	Inc. acc.	Inc. word	Bg. acc.	Bg. word	Neutral
original_ambiguous_written_in	ambiguous	1.72	98.28	0.74	98.40	73.12
control_background	control	0.31	99.69	0.17	99.15	81.78
control_word_oriented	control	0.00	100.00	0.00	100.00	75.13
descriptive_ignore_word	descriptive	10.48	89.52	5.59	87.98	80.64
descriptive_not_background	descriptive	5.41	94.59	2.31	92.66	80.78
very_descriptive_ignore_all_distractors	descriptive	9.75	90.25	5.19	89.46	79.42
very_descriptive_stroop_task	descriptive	16.40	83.60	8.96	80.28	79.75
exact_font_color	exact	5.46	94.54	1.80	94.50	79.78
exact_ink_color	exact	8.28	91.72	5.23	88.96	79.74
exact_letters_color	exact	2.59	97.41	1.46	97.12	78.03
vague_colored_word	vague	0.00	100.00	0.00	99.99	73.32
vague_minimal_color	vague	0.00	100.00	0.05	99.73	75.96

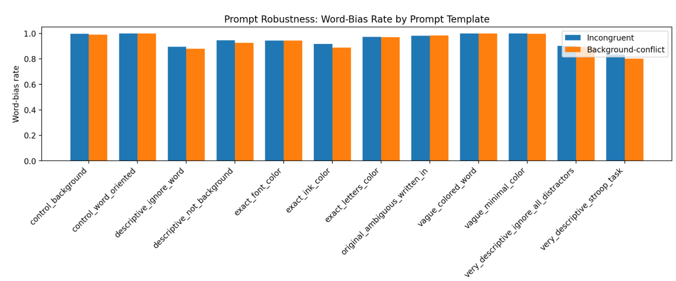

Fig. 2: Prompt robustness experiment. The word-bias rate remains high across vague, exact, descriptive, and control prompt styles. Longer prompts reduce but do not remove the English word-bias effect.

C. Experiment 3: Text-Meaning Recognition Control

The text-meaning control uses black text on a white background. It is not a font-color task. Its purpose is to test whether CLIP can recognize written words in each script without visual color conflict. Table V shows two evaluation modes: `script_match`, where candidate prompts contain the same script as the image, and `english_semantic`, where candidate prompts use English semantic labels.

The English results are 100% for both color words and neutral words in both modes. In contrast, Chinese, Hindi, and Arabic results are very low. This supports the cross-script interpretation: CLIP shows strong Stroop-like word bias mainly when it can process the written script meaning.

TABLE V: Text-meaning recognition control accuracy. Values are percentages.

Language	Script color	Script neutral	Eng. color	Eng. neutral
english	100.00	100.00	100.00	100.00
chinese	3.00	0.52	15.68	0.00
hindi	9.72	0.28	10.00	0.00
arabic	10.00	0.00	10.84	0.00

D. Experiment 4: Multilingual Stroop

The multilingual Stroop experiment uses English, Chinese, Hindi, and Arabic, with the same canonical color labels across scripts. Table VI reports the two conflict conditions, because they are most relevant to interference.

The most important pattern is the contrast between English and the non-English scripts. For English, CLIP shows 91.70% word bias in incongruent images and 88.58% word bias in background-conflict images. For Chinese, Hindi, and Arabic, word bias is close to zero in background-conflict and below 3.3% in incongruent. Errors in non-English background-conflict images shift mostly to background bias: 70.98% for Chinese, 72.50% for Hindi, and 77.60% for Arabic. This means that when CLIP does not read the script as a color word, its failures are dominated by visual/background distraction rather than semantic word interference.

TABLE VI: CLIP ViT-B/16 multilingual Stroop results for conflict conditions. Values are percentages.

Language	Condition	N	Acc.	Word bias	Background bias	Other
arabic	background_conflict	5,000	20.92	0.14	77.60	1.34
arabic	incongruent	5,000	79.14	3.02	9.06	8.78
chinese	background_conflict	5,000	27.62	0.28	70.98	1.12
chinese	incongruent	5,000	80.82	2.08	8.10	9.00
english	background_conflict	5,000	5.12	88.58	6.30	0.00
english	incongruent	5,000	8.30	91.70	0.00	0.00
hindi	background_conflict	5,000	26.80	0.08	72.50	0.62
hindi	incongruent	5,000	74.30	3.26	20.96	1.48

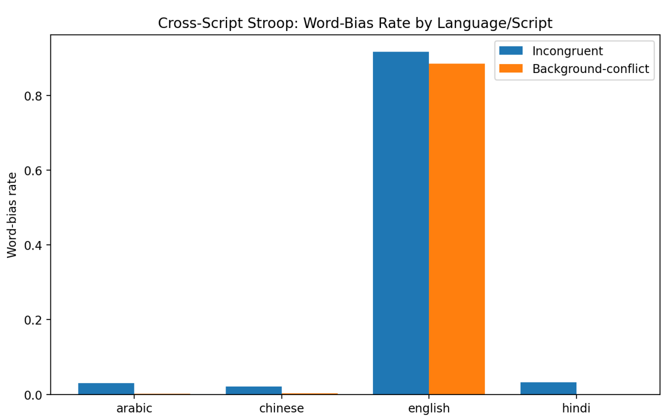

Fig. 3: Cross-script CLIP word-bias rate. English produces strong word bias, while Chinese, Hindi, and Arabic show very low word bias under the same font-color task.

E. Experiment 5: Qwen2.5-VL-3B

Table VII shows Qwen2.5-VL-3B results on the balanced 16,000-image subset. The subset is uniformly distributed across four languages and four conditions, with 1,000 images per language-condition pair. Qwen performs much better than CLIP in most conditions. English incongruent accuracy is 98.00% with only 2.00% word bias, while English background-conflict accuracy is 93.30%. The hardest Qwen cases are background-conflict examples in Arabic, Chinese, and Hindi. These conditions have lower accuracy but still show far less English-style word bias than CLIP.

TABLE VII: Qwen2.5-VL-3B results on the balanced 16,000-image multilingual subset, reporting the two conflict conditions. Values are percentages.

Language	Condition	N	Acc.	Word bias	Background bias	Other
arabic	background_conflict	1,000	69.10	2.90	2.40	25.60
arabic	incongruent	1,000	99.40	0.00	0.00	0.60
chinese	background_conflict	1,000	73.30	14.40	0.60	11.70
chinese	incongruent	1,000	97.80	2.20	0.00	0.00
english	background_conflict	1,000	93.30	4.80	0.00	1.90
english	incongruent	1,000	98.00	2.00	0.00	0.00
hindi	background_conflict	1,000	67.70	3.70	3.20	25.40
hindi	incongruent	1,000	100.00	0.00	0.00	0.00

The Qwen result suggests that instruction-following VLMs can sometimes use explicit task instructions more effectively than contrastive CLIP prompts. However, this should not be over-interpreted as a complete solution. Qwen still has errors in background-conflict conditions, and more model comparisons are needed.

VI. Interpretability Analysis

A. CLIP Grad-CAM-Style Heatmaps

For CLIP, the notebook uses a custom Grad-CAM-style method over `clip_model.visual.conv1`. This is not the official Grad-ECLIP method; it is a custom patch-level attribution method for visualizing which image regions contribute to the CLIP image-text similarity score. For a given image and prompt, the score is backpropagated through the visual encoder, the gradients and activations are combined, and the heatmap is resized to image resolution.

For a heatmap H and binary text mask M_{text}, text energy share is:

$$E_{text} = \frac{\sum_{i \in M_{text}} H_i}{\sum_i H_i}. \quad (6)$$

Text enrichment divides this by the text area share:

$$\text{TextEnrichment} = \frac{E_{text}}{|M_{text}|/|M|}. \quad (7)$$

Values above 1 indicate heatmap concentration on the text region beyond chance area.

Table VIII lists selected CLIP heatmap metrics. The word-bias examples confirm that CLIP's selected prompt corresponds to the written word rather than the correct font color. The heatmaps often overlap the text region, which suggests that the model is not simply looking away from the letters; instead, the representation of the letters aligns more strongly with word meaning than with visual font color.

B. Qwen Attention Heatmaps

For Qwen, the paper does not use Grad-CAM. Instead, it uses attention-based visual token heatmaps. The method measures attention from generated answer tokens to visual image tokens after Qwen produces a color answer. These maps

Fig. 4: CLIP decision heatmaps for word-bias examples. Left: an incongruent English example where the word RED is yellow but CLIP predicts red. Right: a background-conflict example where the word ORANGE is printed in blue on yellow but CLIP predicts orange.

TABLE VIII: Selected CLIP decision heatmap metrics.

Condition	Word	Font	Pred.	Error	Text energy	Text enrichment
incongruent	GREEN	purple	green	word_bias	0.083	0.678
incongruent	YELLOW	green	yellow	word_bias	0.041	0.413
incongruent	WHITE	red	white	word_bias	0.174	1.517
background_conflict	PURPLE	green	purple	word_bias	0.000	0.000
background_conflict	PURPLE	yellow	purple	word_bias	0.071	0.698
background_conflict	PURPLE	brown	purple	word_bias	0.677	6.689
neutral	SHIP	brown	brown	correct_font_color	0.097	0.977
neutral	HOUSE	gray	gray	correct_font_color	0.000	0.000
neutral	CHAIR	green	green	correct_font_color	0.124	1.089
congruent	BLUE	blue	blue	correct_font_color	0.055	0.529
congruent	ORANGE	orange	orange	correct_font_color	0.218	2.045
congruent	BLUE	blue	blue	correct_font_color	0.569	5.439

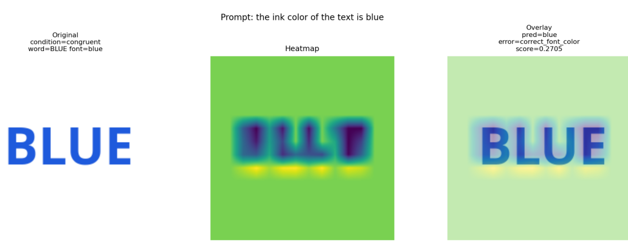

Fig. 5: A congruent CLIP example. Since word meaning and font color agree, this case cannot distinguish reading from seeing, but it verifies that the decision heatmap is concentrated near the text region.

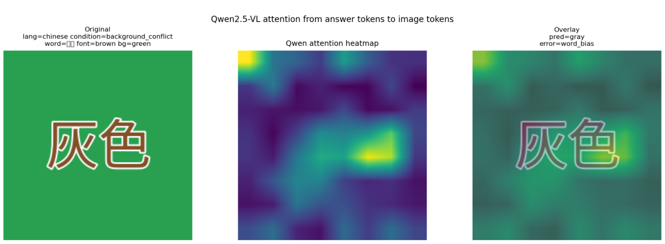

Fig. 6: A representative Qwen2.5-VL attention-based heatmap. The map visualizes attention from generated answer tokens to visual tokens for a Chinese background-conflict example. Unlike the CLIP heatmaps, this visualization is attention-based rather than gradient-based.

are architecture-dependent and should be treated as qualitative diagnostics rather than causal explanations. They are therefore reported separately from CLIP Grad-CAM-style heatmaps.

The Qwen heatmap examples are most useful as a complementary qualitative check. They show that the instruction-following model can attend to the rendered text and surrounding visual field while generating a color answer, but they should not be interpreted as directly equivalent to CLIP similarity attribution. For this reason, the paper uses Qwen heatmaps only as supporting interpretability evidence, while the main quantitative claims are based on accuracy, word-bias rate, background-bias rate, and other-error rate.

VII. Discussion

The results support three main observations. First, CLIP shows a strong English word-bias failure mode. It performs well on congruent and neutral examples but follows the word meaning in conflict examples. This implies that CLIP is capable of visual color recognition, but its representation is strongly pulled toward readable English word meaning when the word is also a color label.

Second, prompt clarity is not sufficient to remove the failure. The descriptive prompts explicitly instruct the model to ignore the word meaning and background, yet word bias remains high. This suggests that the behavior is not merely a poorly worded prompt artifact. It is likely related to how CLIP aligns text-containing images with natural-language descriptions.

Third, the multilingual results show that word bias is script-dependent. English produces high semantic word interference, while Chinese, Hindi, and Arabic produce low word bias under the same font-color task. Combined with the text-meaning control, this suggests that interference is strongest when CLIP can process the written word as a semantic label in the prompt space.

The Qwen results provide an important contrast. Qwen2.5-VL-3B is given a direct instruction to ignore word meaning and background color. It follows this instruction much more reliably than CLIP in most cases. This difference suggests that instruction-following VLMs may partially mitigate the CLIP-style word bias. However, background-conflict cases remain harder, particularly for Arabic, Chinese, and Hindi, so the issue is not completely solved.

VIII. Limitations

The benchmark is intentionally synthetic so that word meaning, font color, and background color can be controlled independently. This controlled design is a strength for isolating cue conflict, but it also defines the scope of the claims: the results should be interpreted as diagnostic evidence about controlled cue competition rather than as a full measurement of all real-world image-text settings. Real-world images may contain irregular typography, lighting variation, occlusion, perspective distortion, and contextual objects that are not fully covered by the current generator.

The label space is restricted to ten canonical color names. This makes error categorization precise and reproducible, but it does not address fine-grained color perception, ambiguous shades, or culturally variable color naming. The multilingual evaluation is also limited to four scripts. English, Chinese, Hindi, and Arabic provide a useful initial comparison, but they do not exhaust the range of writing systems, fonts, and script-specific visual properties.

The interpretability results should be read as qualitative support for the behavioral findings. The CLIP heatmaps are custom Grad-CAM-style attributions over the visual encoder, and Qwen heatmaps are attention-based visual-token maps. These methods provide insight into model behavior, but they are not interchangeable and should be complemented by intervention tests before making strong causal claims about internal mechanisms.

IX. Future Work

Future work should expand the benchmark in three directions. First, model coverage can be increased by evaluating additional contrastive and instruction-following VLMs, including OpenCLIP, SigLIP, BLIP-style models, and larger Qwen variants (Cherti et al., 2023; Zhai et al., 2023; Li et al., 2023). This would help determine whether the observed English word-bias pattern is specific to CLIP or shared across model families.

Second, the benchmark can be made more visually diverse while preserving controlled labels. Useful extensions include additional fonts, font weights, rotations, blur, shadows, texture, compression artifacts, and naturalistic backgrounds. These perturbations would test whether the same cue-conflict pattern persists under more realistic visual conditions.

Third, future analysis should include intervention-based tests. For example, masking the word region, masking the background, replacing the script while keeping font color fixed, or swapping the word while preserving the same visual color would allow stronger causal claims about which cue drives each prediction. A further extension is conflict-aware training or prompt-tuning, where models are trained or adapted to separate visual attributes from written semantic distractors.

X. Conclusion

This paper presents an expanded multilingual Stroop benchmark for testing visual-textual cue interference in VLMs. CLIP ViT-B/16 shows strong English word bias, and this bias persists even under exact and descriptive prompt templates. The effect is script-dependent: English produces strong word-bias interference, while Chinese, Hindi, and Arabic largely shift errors toward background or other visual confusions. The text-meaning control supports this interpretation because CLIP recognizes English text reliably but performs poorly on the non-English scripts. Qwen2.5-VL-3B performs much better on the balanced subset, suggesting that instruction-following VLMs may better separate visual font color from written word meaning. Overall, the benchmark shows that a simple Stroop-style setup can expose whether a multimodal model is reading, seeing, or confusing the two.

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APPENDICES

Appendix A: Prompt Templates

The prompt robustness experiment reused the English Stroop images with multiple prompt styles to test whether word bias was only caused by vague wording.

- **Vague:** minimal prompts such as `{color}` or `a {color} word`.
- **Ambiguous:** original wording such as `the text is written in {color}`.
- **Exact:** direct visual prompts such as `the ink color of the text is {color}` and `the font color is {color}`.
- **Descriptive:** longer prompts instructing the model to ignore word meaning and background color.
- **Control:** word-oriented and background-oriented prompts such as `a photo of the word ``{color}''` and `the background color is {color}`.

Appendix B: Reproducibility Artefacts

The project was implemented in Google Colab using Python. The main artefacts include:

- generated English, multilingual, and text-meaning control image datasets;
- metadata CSV files with image path, language, condition, word, font color, background color, and label;
- CLIP prediction and summary CSV files for all experiments;
- Qwen2.5-VL prediction and summary CSV files for the balanced subset;
- prompt robustness and cross-script word-bias plots;
- CLIP Grad-CAM-style heatmaps and Qwen attention heatmaps.

In all Stroop-style experiments, the ground-truth label is the visible font color. The written word and background color are treated as distractor cues.

Appendix C: Generative AI Statement

Generative AI assistance was used for drafting support, that is for aligning images and text, code-structuring guidance, explanation, and writing refinement; organising interpretations, and final academic responsibility remain with the author.